Functions

CSE 101 Section 03

Introduction to Algorithms and Computing

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This program is one long, complex sequence of statements

```
statement
statement
statement
statement
statement
statement
statement
statement
statement
statement
statement
statement
```

In this program, the task has been divided into smaller tasks, each of which is performed by a separate function

```
function function1()
        statement
        statement
        statement
end
function function2()
        statement
        statement
        statement
end
function function3()
        statement
        statement
        statement
end
function function4()
        statement
        statement
        statement
end
```

What are functions?

A grouping of statements together to perform some functionality.

Benefits

- Readability
- Reusability
- Modularity (i.e., breaking program into smaller tasks)

Anatomy of a function

MATLAB keywords

function name

outputs

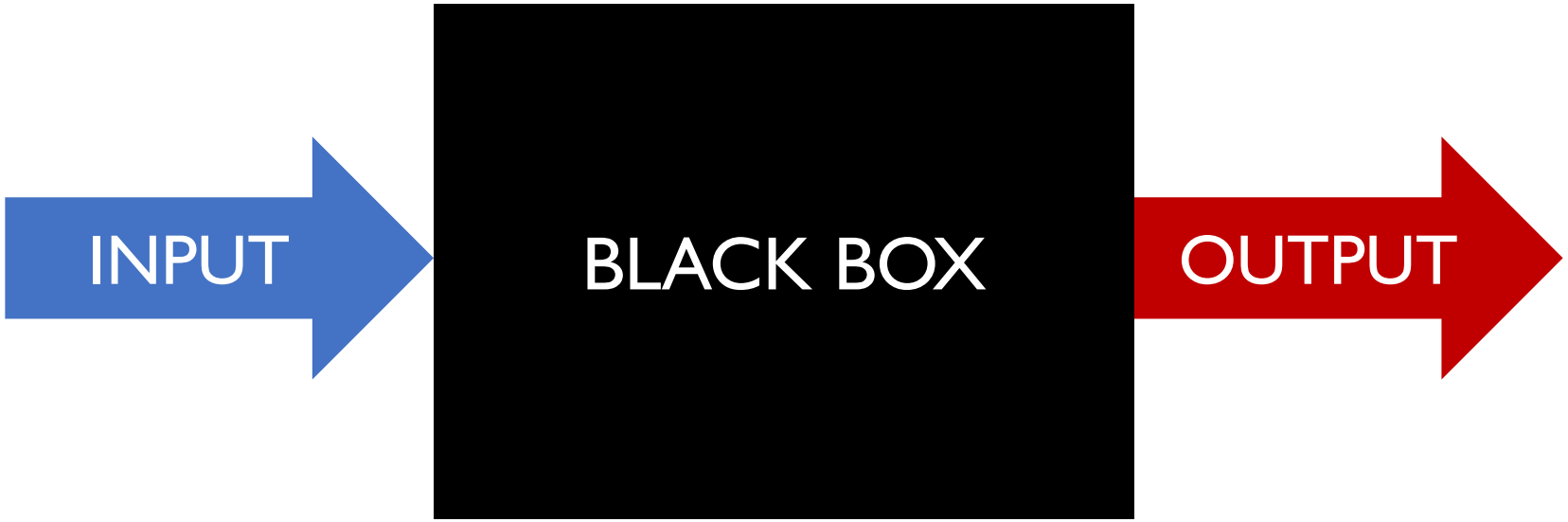
```
function [y1,..,yN] = myfun(x1,..,xM)
    statement(s)
end
```

inputs

The name of the function should reflect what it does. Same naming rules as for variables.

```
Examples: findAverage()
          printReport()
```

Functions



Example

```
main()

function singHappyBD()
     disp("Happy birthday to you")
     disp("Happy birthday to you")
     disp("Happy birthday to you, Random Person")
     disp("Happy birthday to you!")
end


function main()
     singHappyBD()
     singHappyBD()
end
```

Few rules to note about MATLAB functions

- In a Matlab script file which contains commands and function definitions, functions must be at the end of the file.

- Script files cannot have the same name as a function in the file.

- But in a function file which contains only function definitions. The name of the file must match the name of the first function in the file.

- Read more here: https://www.mathworks.com/help/matlab/ref/function.html

Example – with an input

```
main()

function singHappyBD(person)
     disp("Happy birthday to you")
     disp("Happy birthday to you")
     disp("Happy birthday to you, " + person)
     disp("Happy birthday to you!")
end

function main()
     singHappyBD("Gracie")
     name = "Tulie";
     singHappyBD(name)
end
```

Another Example – Not a good one!

```
distanceBetweenPoints()

function distanceBetweenPoints()
    x1 = input("Enter x1: ");
    y1 = input("Enter y1: ");
    x2 = input("Enter x2: ");
    y2 = input("Enter y2: ");
    distance = sqrt((x2-x1)^2 + (y2-y1)^2);
    disp(distance)
end
```

Why is that not a good function?

- Not general!
- Doing more than one thing – getting input from the user, doing the calculation and printing the result on the screen.
- Requires input from the user (thus not generally reusable)
- Output just printed on the screen
  - cannot be used by the program in any way later

More general!

```
function distance = distanceBetweenPoints(x1, x2, y1, y2)
     distance = sqrt((x2-x1)^2 + (y2-y1)^2);
end
```

- How would we get the values x1, x2, y1, y2?
  - From the user OUTSIDE the function
  - Read from an external source
  - Generated by the program somehow...

- Does only one thing – finds the distance between two points. If user input is required or the result needs to be printed, do it elsewhere!

How does the function get the values?

```
dis = distanceBetweenPoints(20, 40, 100, 300)
```

```
function distance = distanceBetweenPoints(x1, x2, y1, y2)
    distance = sqrt((x2-x1)^2 + (y2-y1)^2);
end
```

What can the inputs (aka arguments) look like?

```
dis = distanceBetweenPoints(20, 40, 100, 300)

dis = distanceBetweenPoints(firstx, secondx, firsty,
secondy) % These need values assigned

dis = distanceBetweenPoints(getPoint(), getPoint(),
getPoint(), getPoint()) % where getPoint() is a function
that returns an integer

function distance = distanceBetweenPoints(x1, x2, y1, y2)
    distance = sqrt((x2-x1)^2 + (y2-y1)^2);
end
```

Local Variables

```
function distance = distanceBetweenPoints(x1, x2, y1, y2)
     distance = sqrt((x2-x1)^2 + (y2-y1)^2);
end
```

The variable "distance" is a local variable.
It is *NOT visible anywhere outside the function.*
SO ... how do we get the result from this function?

```
dis = distanceBetweenPoints(20, 40, 100, 300)
```

Store it in a new variable

More on Local Variables

```
x = function1();
disp(x) %what is printed?
x = function2();
disp(x) %what is printed?? OOPS!

function x = function1()
      x = 5;
end

function function2()
      x = "elephant";
end

function function3()
      x = 17;
end
```

Because variables are defined inside functions, we can have many with the same names. They are NOT the same variables!!!

More on Outputs (aka Return Values)

```
fact = factorial(5);
disp(fact);
disp(factorial(5));
result = factorial(2) + factorial(3);
disp(result);

function fact = factorial(n)
      if n==0
            fact = 1;
      else
            fact = 1;
            for i = [1:n]
                  fact = fact .* i;
            end
      end
end
```

Functions with no output (or return value)

```
disp(singHappyBD()) % is this correct?
singHappyBD() % or is this?
```

```
function singHappyBD()
    disp("Happy birthday to you!!")
end
```

Returning multiple values

```
[theSum, theAvg] = foo(40, 60, 50)
```

```
function [total, average] = foo(mark1,
mark2, mark3)
    total = mark1 + mark2 + mark3;
    average = total/3;
end
```

Next…

Input / Output